low approximation error, vAttention supports accurate long-form generation, producing sequences of up to 32K tokens while matching full-attention accuracy on AIME(Maxwell-Jia, 2024). Additionally, we show that there is a near-perfect correlation between the user-specified tolerance ϵ and the average empirically observed error in attention outputs, showcasing the effectiveness of vAttention in exposing the quality–efficiency trade-off of sparse attention to the end user (see Figure 1).

2 RELATED WORK

Existing work on sparse attention can be categorized into the following types, covering early explorations to recent efforts. Detailed related work is presented in Appendix B.

Static Sparse Attention and KV Cache compression methods Early sparse attention methods used fixed patterns to limit tokens during decoding. For instance, StreamingLLM (Xiao et al., 2023) attends to "attention sinks" (early tokens) and a sliding window of recent tokens. Later work (Zhang et al., 2023; Xiao et al., 2024) showed that such static patterns fail to generalize, motivating dynamic sparsity. StreamingLLM's key insight—that sinks and local windows are essential—remains central to subsequent methods. Another direction compresses the KV cache by discarding tokens, as in ScissorHands (Liu et al., 2024b), H2O (Zhang et al., 2023), FastGen (Ge et al., 2023), and SnapKV (Li et al., 2024a). While memory-efficient, these approaches lack generality, since irreversible pruning struggles in settings like multi-turn dialogue, where token relevance shifts across turns.

Approximate top-k based Sparse Attention A class of sparse attention methods approximates top-k token selection—identifying tokens with the highest query–key inner products—since exact computation is $O(nd)$ and undermines efficiency. Examples include Double Sparsity (Yang et al., 2015), which sparsifies partial channels; Loki (Singhania et al., 2024), which applies low-rank decomposition; and InfLLM (Xiao et al., 2024) and Quest (Tang et al., 2024), which use page-level approximations. As top-k identification is essentially an inner product search problem (Desai et al., 2025), many methods adapt approximate nearest neighbor (ANN) techniques: PQCache (Zhang et al., 2025) leverages product quantization, SqueezeAttention (Hooper et al., 2024) employs hierarchical clustering, Retrieval Attention (Li et al., 2024b) adopts graph-based ANN search, and HashAttention (Desai et al., 2025) encodes queries and keys as bit signatures. These approaches improve scalability by narrowing the search to promising tokens, but their dependence on oracle top-k selection imposes a fundamental limitation. As demonstrated in MagicPig (Chen et al., 2024) and further analyzed here, even access to the exact top-k tokens under full attention does not always suffice to approximate the original output, highlighting the need to move beyond top-k selection in sparse attention design.

Approximate Top-P based Sparse Attention A key limitation of top-k–based sparse attention is that a fixed sparsity level fails to generalize across modules. To address this, recent work adopts Top-p coverage, selecting a variable number of tokens whose cumulative attention scores exceed a threshold p, thereby adapting to varying importance distributions while offering error control. Exact Top-p, however, is even costlier than top-k as it requires sorting or aggregating all scores; methods therefore approximate coverage—for example, Tactic (Zhu et al., 2025) models attention decay with a power-law distribution to estimate the required number of tokens. As we show, Top-p is not the most efficient way to achieve a target error bound; more principled mechanisms, including vAttention, attain comparable or better accuracy with fewer tokens.

MagicPig: LSH sampling-based Sparse Attention MagicPig (Chen et al., 2024) was among the first to highlight the issues with top-k–based sparse attention. It employs Locality Sensitive Hashing (LSH)(Gionis et al., 1999) to select tokens for attention computation. While LSH is suboptimal for approximate nearest neighbor (ANN) search due to its data-agnostic projections, in this context, it provides a principled sampling-based mechanism for approximating attention (Luo & Shrivastava, 2018). Tokens retrieved via LSH have associated probabilities reflecting how they were sampled under the randomized construction of the LSH table. Early exploration of vAttention was inspired by MagicPig, and we elaborate further on the attention computation in subsequent sections. Another related work, SampleAttention (Zhu et al., 2024), samples structured patterns to approximate attention.

Existing methods lack guarantees on the quality of approximation. In contrast, vAttention offers explicit control over approximation quality, providing both reliability and flexibility.